

Mock Final Exam #2 – Machine Learning (MDS + BDMA + MIRI)

Duration: 1h30m

All questions have equal weight. Justify all your answers. Be concise and clear.

Question 1

State whether the following statements are **True** or **False**, and justify each answer briefly:

- (a) Feature scaling is required before training a decision tree.
- (b) Increasing the regularization strength in ridge regression increases bias.
- (c) Bagging reduces the variance of base models.
- (d) k-NN requires feature scaling to behave properly.
- (e) Logistic regression always results in a linear decision boundary.
- (f) The output activation function of a neural network is determined by the type of task (e.g., regression or classification).

Question 2

Compare and contrast the following pairs of models in terms of:

- Sensitivity to irrelevant features
 - Interpretability
 - Scalability with large feature dimensions
- (a) Logistic Regression vs. Decision Tree
 - (b) Random Forest vs. Multilayer Perceptron (MLP)

Question 3

You are tasked with estimating the generalization error of a model trained on a small dataset (100 samples). You consider two alternatives:

- (a) 10-fold cross-validation
- (b) Leave-one-out cross-validation (LOOCV)

Compare the two in terms of:

- Bias and variance of the error estimate
- Computational cost
- Suitability for model selection

Question 4

Consider a scenario where your model overfits the training data. For each of the following models, propose a way to reduce overfitting and briefly justify it:

- (a) Decision Tree
- (b) k-NN
- (c) MLP Neural Network
- (d) Logistic Regression

Question 5

Design a 2D dataset (two input features) where:

- Logistic regression fails to classify the classes correctly
- A Random Forest achieves perfect classification

You may describe the data, draw the points, or specify the rule behind class separation. Explain why the performance differs between the models.